

Ariel Trejo

Email: arieltrejo0804@gmail.com

Phone Number: 956-261-0195



EDUCATION

The University of Texas Rio Grande Valley, Edinburg, TX

B.S. in Computer Science, GPA: 3.77/4.0

Publications: [Enhancing Immersive Experiences: Computer Vision Enhanced Mixed Reality in Smart Manufacturing](#)

WORK EXPERIENCE

EOS Additive Manufacturing, Pflugerville, TX

Returning Data Analysis Intern

May 2025 – Aug 2025

- Migrated the company's static HTML database interface into a full React.js dashboard with dynamic, real-time data interaction.
- Implemented interactive visualization pipelines using D3.js and Recharts; upgraded tables to MUI DataGrid for high-volume performance.
- Deployed the platform on an Azure Virtual Machine and configured firewall rules, access controls, and IP restrictions.
- Researched AI-driven database querying using LangChain and Ollama for potential LLM integration.

Data Analysis Intern

June 2024 – Aug 2024

- Built a full-stack data management system processing thousands of manufacturing records.
- Configured and connected a PostgreSQL database to a Python-based data pipeline, including data type correction and schema cleanup for reliable querying.
- Developed backend services using Python + Flask; built a responsive frontend in HTML, JavaScript, and Bootstrap.

UTRGV Institute for Advanced Manufacturing, Research Assistant, Edinburg, TX

Oct 2023 – Jan 2026

Digital Twin Research Project

- Developed a Digital Twin which consists of augmented reality using Microsoft HoloLens to produce a 3-D model of a KUKA AG which is interactable for the user, utilizing Unity, C# and Vuforia.
- Created an Azure SQL Database to connect with HoloLens, enabling users to interact with a digital map and retrieve object information.

Hydrocat Autonomous Boat Research Project

- Configured Nvidia Jetson Nano (Ubuntu + CUDA + Conda) for ML inference and real-time data processing.
- Developed a Python-based YOLO object detection system with threshold-based “red-zone” auto-stop logic.
- Built a secure networking pipeline to stream live telemetry and camera feed over a private local network.
- Experienced with Mission Planner for GPS, compass, and sensor configuration.

TECHNICAL SKILLS

Programming Languages: Python, C++, C, C#, SQL, PostgreSQL, JavaScript, HTML

Technology, Tools, & Frameworks: React.js (React), Microsoft Azure (Azure), Azure Virtual Machines, SQLAlchemy, Unity, Flask Framework, LangChain, Ollama, D3.js, Recharts, JSON, Bootstrap, Network Security / Firewall Configuration, Large Language Models (LLMs), Generative AI, Natural Language Processing (NLP), Digital Twin, Augmented Reality (AR), Robotics